
Degree-Aware Curriculum Hard-Negative Sampling for LightGCL

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Abstract

LightGCL (ICLR 2023) is a state-of-the-art graph contrastive learning model for recommendation that contrasts a LightGCN view of the user-item graph with an SVD-reconstructed view, yet its ranking objective still relies on *uniformly* sampled negative items in the BPR loss. We propose a **degree-aware curriculum hard-negative sampler** for this BPR component: negatives are drawn from a mixture of a uniform distribution and a Personalized-PageRank (PPR) hard-negative distribution, whose mixture weight grows with training progress (curriculum) and with each user’s activity level (degree), so that sparse users keep mostly random negatives, limiting false-negative bias. We evaluate on Yelp2018 (Gowalla time-permitting) with Recall@20 and NDCG@20 against MF-BPR, LightGCN, SimGCL, and LightGCL, including a sampling-strategy ablation and a user-degree-bucket breakdown.

1 Application domain: dataset, task, and metrics

Dataset. Our primary dataset is **Yelp2018**, a standard implicit-feedback recommendation benchmark of user check-ins at local businesses; if time permits we additionally run **Gowalla** (user-location check-ins). Both datasets, with their public train/test splits, ship with the official LightGCL repository,¹ so no extra preprocessing is required. Table 1 reports the statistics of the standard benchmark splits [2].

Table 1: Statistics of the benchmark datasets (standard splits).

Dataset	#Users	#Items	#Interactions	Density
Yelp2018 (primary)	31,668	38,048	1,561,406	0.130%
Gowalla (time-permitting)	29,858	40,981	1,027,370	0.084%

Task and metrics. The task is **top- K recommendation from implicit feedback**: given the bipartite user-item interaction graph, rank all items a user has not interacted with and recommend the top K . Following the standard full-ranking protocol, we report **Recall@20** and **NDCG@20** averaged over test users. In addition, we break both metrics down by *user-degree bucket* (head/mid/tail by

¹<https://github.com/HKUDS/LightGCL>

number of training interactions) to measure long-tail robustness. Since the LightGCL paper already analyzes sparsity and popularity robustness [4], we treat this breakdown as a diagnostic rather than a novelty claim.

Why this dataset. (i) Yelp2018 and Gowalla are the canonical benchmarks for the exact model family we build on (LightGCN, SimGCL, LightGCL), enabling direct comparison with published numbers; (ii) data and code come from the authors’ official repository, making reproduction feasible within the course timeline; (iii) both graphs are sparse with heavily long-tailed user activity, precisely the setting where the choice of BPR negatives—and our degree-aware design—should matter most.

2 Graph ML techniques

2.1 Base model: LightGCN encoder and BPR objective

All compared models embed users u and items i as vectors $e_u, e_i \in \mathbb{R}^d$ and score with the inner product $\hat{y}_{ui} = e_u^\top e_i$. LightGCN [2] refines the embeddings by L layers of linear message passing over the normalized bipartite graph:

$$e_u^{(l+1)} = \sum_{i \in \mathcal{N}_u} \frac{1}{\sqrt{|\mathcal{N}_u| |\mathcal{N}_i|}} e_i^{(l)}, \quad e_i^{(l+1)} = \sum_{u \in \mathcal{N}_i} \frac{1}{\sqrt{|\mathcal{N}_i| |\mathcal{N}_u|}} e_u^{(l)}, \quad e_u = \frac{1}{L+1} \sum_{l=0}^L e_u^{(l)}. \quad (1)$$

Training uses the pairwise BPR loss [1]: for each observed pair (u, i) a negative item j is *sampled* and the model is pushed to rank i above j ,

$$\mathcal{L}_{\text{bpr}} = \sum_{(u, i) \in \mathcal{O}} \mathbb{E}_{j \sim P(\cdot | u)} [-\log \sigma(\hat{y}_{ui} - \hat{y}_{uj})], \quad (2)$$

where $P(\cdot | u)$ is uniform over non-interacted items in all baselines. **This sampler is the single component we modify.**

2.2 LightGCL: SVD-guided graph contrastive learning

LightGCL [4] augments LightGCN with a second view built from a truncated rank- q SVD of the normalized adjacency matrix $\tilde{\mathcal{A}} \approx U_q S_q V_q^\top = \hat{\mathcal{A}}$; propagating embeddings over the low-rank reconstruction $\hat{\mathcal{A}}$ yields view embeddings z'_u, z'_i that emphasize global collaborative structure. The two views are aligned with an InfoNCE loss (shown for users; items are symmetric):

$$\mathcal{L}_{\text{cl}} = \sum_{u \in \mathcal{B}} -\log \frac{\exp(\cos(z_u, z'_u)/\tau)}{\sum_{v \in \mathcal{B}} \exp(\cos(z_u, z'_v)/\tau)}, \quad \mathcal{L} = \mathcal{L}_{\text{bpr}} + \lambda_1 \mathcal{L}_{\text{cl}} + \lambda_2 \|\Theta\|_2^2. \quad (3)$$

Note that LightGCL uses negatives in *two distinct places*: the contrastive loss (3) uses the other in-batch users/items, while the BPR loss (2) uses one explicitly sampled negative item. We intervene *only* in the BPR negative sampler and leave the contrastive component untouched (Figure 1).

2.3 Proposed: degree-aware curriculum hard-negative sampling

Research question: does a hard-negative curriculum modulated by user activity improve the recommendation accuracy and long-tail robustness of LightGCL without increasing false-negative bias?

Hard-negative candidates via PPR. Following PinSAGE [3], we rank all non-interacted items for each user by Personalized PageRank (PPR) on the interaction graph and treat items that are *related but not top-ranked* as hard negatives. PinSAGE’s absolute rank band (2000–5000) was tuned for the web-scale Pinterest graph, so we instead define the band by *percentiles* of each user’s PPR ranking over the item set $\mathcal{I}$:

$$\mathcal{B}_u = \{j \notin \mathcal{N}_u : \text{rank}_{\text{PPR}}(j | u) \in [p_{\text{lo}}|\mathcal{I}|, p_{\text{hi}}|\mathcal{I}|]\}, \quad P_{\text{PPR}}(j | u) = \text{Unif}(\mathcal{B}_u), \quad (4)$$

with $(p_{\text{lo}}, p_{\text{hi}})$ tuned on validation data (e.g., the 5%–15% most PPR-similar items, excluding the very top of the ranking to avoid unobserved positives). PPR scores are precomputed once per dataset.

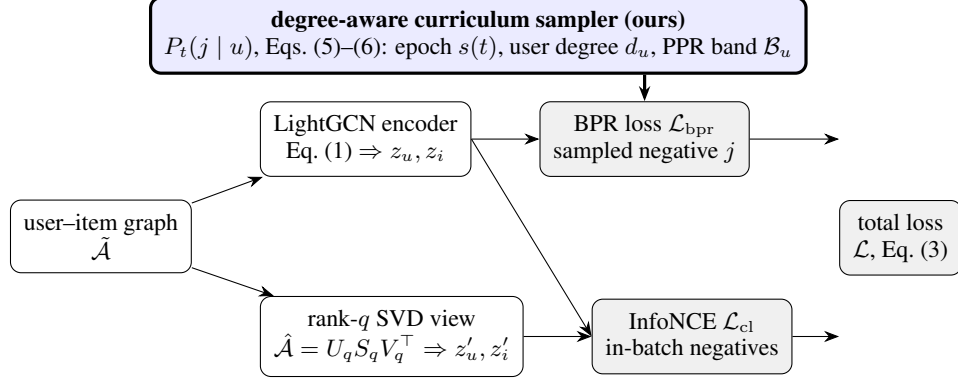


Figure 1: LightGCL with the proposed sampler: both views and the contrastive loss are unchanged; only the uniform BPR negative sampler is replaced (bold).

Degree-aware curriculum mixture. At training epoch t , the BPR negative for user u is drawn from

$$P_t(j | u) = (1 - \alpha_{u,t}) P_{\text{uniform}}(j) + \alpha_{u,t} P_{\text{PPR}}(j | u), \quad (5)$$

where the gate combines a curriculum schedule with the user’s degree d_u :

$$\alpha_{u,t} = s(t) \cdot \sigma(a \log(1 + d_u) + b), \quad s(t) = \min(1, t/T_w). \quad (6)$$

Here $s(t)$ ramps up hard negatives only after a warm-up of T_w epochs (curriculum learning [6]), and the sigmoid gives *active* users harder negatives while *sparse* users keep mostly uniform negatives: for a low-degree user, a PPR-similar item is likely an unobserved positive, so treating it as a negative injects false-negative bias exactly where the model has the least evidence.

Secondary ablation: energy-based SVD rank selection. The rank q of the SVD view is a sensitive hyperparameter of LightGCL (the paper shows very small q , around 5, can suffice [4]). As a *minor* ablation—a hyperparameter-selection heuristic, not a model contribution—we test choosing q by the spectral energy-ratio rule $q^* = \min\{q : \sum_{i \leq q} \sigma_i^2 / \sum_j \sigma_j^2 \geq \rho\}$ with $\rho \in \{0.6, 0.8\}$, compared against the repository default and validation-tuned q .

Why these models fit the data. Yelp2018/Gowalla are ID-only bipartite implicit-feedback graphs, exactly the setting LightGCN’s featureless linear propagation targets, and LightGCL’s SVD view is cheap at this scale while robust to sparsity. The long-tailed degree distribution motivates a *degree-conditioned* sampler rather than a one-size-fits-all hard-negative scheme. Since the data carry no metadata, we deliberately frame low-degree users as *long-tail/sparse* users, not “cold-start” (which would require side information for entirely new nodes).

3 Experimental plan

Baselines and protocol. We compare against MF-BPR [1], LightGCN [2], SimGCL [5], and LightGCL [4], using the official LightGCL code and hyperparameters where available. Primary experiments run on Yelp2018; Gowalla only if time remains. Single-seed runs screen configurations (Table 2); the final configurations (baselines, S0, and the best of S1–S3) are re-run with **three seeds** and reported as mean $\pm$ std of Recall@20/NDG@20, plus the per-degree-bucket breakdown (Section 1). The energy-based rank rule is a small side ablation on LightGCL only.

Feasibility and timeline (report due July 17). Day 1: environment smoke test (Python 3.9 / PyTorch 1.12 / CUDA 11.3) and reproduction of LightGCL on Yelp2018. Days 2–3: PPR precomputation and sampler implementation (S1–S3). Days 4–5: single-seed screening and degree-bucket analysis. Days 6–7: three-seed final runs and report writing. The single main contribution is deliberately scoped to be completable and verifiable within one week.

Table 2: Ablation over BPR negative-sampling strategies (all on LightGCL).

ID	Sampler	Curriculum $s(t)$	Degree gate
S0	uniform random (default)	–	–
S1	PPR hard negatives, Eq. (4)	–	–
S2	mixture, Eq. (5)	fixed ramp	off ($\alpha_{u,t} = s(t) \cdot \bar{\alpha}$)
S3	mixture, Eq. (5)	fixed ramp	on, Eq. (6)

References

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